

DAY-10

Weekly Review and Practice Practical (2 hours): Create a network with multiple VLANs and configure inter-VLAN routing. Task: Submit a report on VLAN and inter-VLAN routing setup.

1. On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.

ANS :

➤ **Create a network containing:**

- 3 switches
- 6 PCs — 2 PCs connected to each switch
- VLAN 10 → **Sales**
- VLAN 20 → **Support**

➤ **Device connections**

Device	Port	Connected To	VLAN
PC1	NIC	SW1 Fa0/1	VLAN 10 Sales
PC2	NIC	SW1 Fa0/2	VLAN 20 Support
PC3	NIC	SW2 Fa0/1	VLAN 10 Sales
PC4	NIC	SW2 Fa0/2	VLAN 20 Support
PC5	NIC	SW3 Fa0/1	VLAN 10 Sales
PC6	NIC	SW3 Fa0/2	VLAN 20 Support
SW1	Fa0/23	SW2 Fa0/23	Trunk
SW2	Fa0/24	SW3 Fa0/23	Trunk
R1	G0/0	SW1 Fa0/24	Trunk

➤ **IP Addressing**

VLAN 10 — Sales

Network: 192.168.10.0/24

Gateway:

192.168.10.1

PC	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC3	192.168.10.12	255.255.255.0	192.168.10.1
PC5	192.168.10.13	255.255.255.0	192.168.10.1

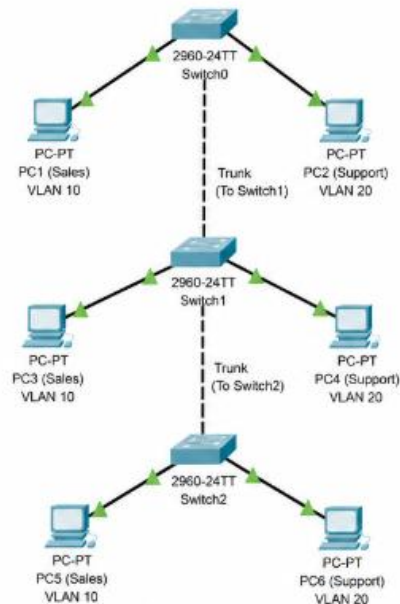
VLAN 20 — Support

Network: 192.168.20.0/24

Gateway:

192.168.20.1

PC	IP Address	Subnet Mask	Default Gateway
PC2	192.168.20.11	255.255.255.0	192.168.20.1
PC4	192.168.20.12	255.255.255.0	192.168.20.1
PC6	192.168.20.13	255.255.255.0	192.168.20.1



2. Configure trunk ports between the switches to allow VLAN traffic to pass through. Verify trunking by using the show interfaces trunk command and submit a screenshot of the output.

ANS :

SW1 → SW2 trunk

On **SW1**:

enable

configure terminal

interface fa0/23

switchport mode trunk

switchport trunk allowed vlan 10,20

exit

interface fa0/24

switchport mode trunk

switchport trunk allowed vlan 10,20

exit

end

write memory

Here:

- Fa0/23 → SW2
- Fa0/24 → Router R1

SW2 → SW1 and SW3 trunks

On **SW2**:

enable

configure terminal

interface fa0/23

switchport mode trunk

```
switchport trunk allowed vlan 10,20
```

```
exit
```

```
interface fa0/24
```

```
switchport mode trunk
```

```
switchport trunk allowed vlan 10,20
```

```
exit
```

```
end
```

```
write memory
```

Here:

- Fa0/23 → SW1
- Fa0/24 → SW3

SW3 → SW2 trunk

On **SW3**:

```
enable
```

```
configure terminal
```

```
interface fa0/23
```

```
switchport mode trunk
```

```
switchport trunk allowed vlan 10,20
```

```
exit
```

```
end
```

```
write memory
```

Verify Trunking

Run this command on each switch:

```
show interfaces trunk
```

3. Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter-VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.

ANS :

Router configuration

Connect:

R1 G0/0 ↔ SW1 Fa0/24

Open **R1** → **CLI**.

First configure the physical interface:

enable

configure terminal

interface gigabitEthernet 0/0

no shutdown

exit

Now configure VLAN 10:

interface gigabitEthernet 0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

exit

Configure VLAN 20:

interface gigabitEthernet 0/0.20

encapsulation dot1Q 20

ip address 192.168.20.1 255.255.255.0

exit

Finally:

end

write memory

Router Configuration — Complete Version

You can also enter everything together:

```
enable
```

```
configure terminal
```

```
interface gigabitEthernet 0/0
```

```
no shutdown
```

```
exit
```

```
interface gigabitEthernet 0/0.10
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
exit
```

```
interface gigabitEthernet 0/0.20
```

```
encapsulation dot1Q 20
```

```
ip address 192.168.20.1 255.255.255.0
```

```
exit
```

```
end
```

```
write memory
```

Verify Router Configuration

Run:

```
show ip interface brief
```

You can also check:

```
show running-config
```

Configure PC IP Addresses

PC → Desktop → IP Configuration

PC1

IP Address: 192.168.10.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

PC2

IP Address: 192.168.20.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

PC3

IP Address: 192.168.10.12

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

PC4

IP Address: 192.168.20.12

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

PC5

IP Address: 192.168.10.13

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

PC6

IP Address: 192.168.20.13

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

4. Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works. Hint : Check sub-interface IPs, trunk configuration, and default gateways on PCs.

ANS :

PC1 → PC2

PC1 belongs to VLAN 10:

192.168.10.11

PC2 belongs to VLAN 20:

192.168.20.11

PC1 :

Desktop → Command Prompt

Enter:

ping 192.168.20.11

Expected result:

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Therefore, **inter-VLAN communication is successful.**

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=126
Reply from 192.168.20.11: bytes=32 time<1ms TTL=126
Reply from 192.168.20.11: bytes=32 time<1ms TTL=126
Reply from 192.168.20.11: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```